Andrew Li

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Education

Georgia Institute of Technology — Atlanta, GA
B.S./M.S. in Computer Science, Undergrad GPA: 3.74 — *Aug. 2023 – May 2027*

- **Coursework:** Machine Learning, Computer Vision, Perception and Robotics, Intro to AI, Linear Algebra, DSA

Experience

Sandia National Laboratories — Albuquerque, NM (On-site → Remote)
Year-Round AI for Autonomy Intern — *May 2025 – Present*

- Developing modular C++ firmware for a platform-agnostic microcontroller-based drone flight logging module to ensure high-frequency capture of inertial, barometric, and GPS data
- Engineered a semantic navigation stack in simulation for a Spot robot dog, utilizing Gemma 3 locally via Ollama for contextual reasoning and behavior trees to orchestrate high-level planning
- Developed a custom C++ ROS2 interface for Kromek GR1+ spectrometers and SparkFun GPS, leading Jackal UGV field trials for real-time radiation mapping as detailed in report SAND2025-12871R

Lunar Lab @ Georgia Tech — Atlanta, GA
Research Assistant — *Apr. 2026 – Present*

- Building a real2sim2real pipeline with NKSR and 3D Gaussian Splatting to scale high-fidelity scene reconstruction from typical small-room scenes to large environments, facilitating warehouse robot policy training pipelines
- Capturing multi-modal data (LiDAR, dual fisheye, RGB-D) via a handheld GeoScan S1, and tagging reconstructed meshes with material parameters (Young's modulus, friction) for manipulation-ready scenes

Humanoid Robotics @ Georgia Tech — Atlanta, GA
Software Lead — *Jan. 2026 – Present*

- Leading a team of 5 developers implementing PPO neural network policies for dynamic bipedal locomotion, balance recovery, and soccer dribbling, with a standardized Docker-based Ubuntu workflow for cross-platform reproducibility

Georgia Tech RoboWrestling — Atlanta, GA
Software Mentor — *Sep. 2023 – Feb. 2025*

- Developed embedded C++ firmware on Teensy 4.1 microcontrollers for autonomous opponent detection
- Mentored 15+ students on 500g robot sumo teams from design to competition

Flowers Invention Studio Makerspace — Atlanta, GA
Prototyping Instructor — *Mar. 2024 – Feb. 2025*

- Supervised and taught safe operation of machinery for a community of 1,000+ users e.g., waterjets and 3D printers

Boston University Morphable Biorobotics Lab — Boston, MA
Research Assistant — *Jun. 2022 – Aug. 2022*

- Designed a soft-robotic pick-and-place manipulator and prototyped minimally invasive cardiac devices

Projects

Basketball Rebounding Robot — *Sep. 2025 – Present*

- Leveraged ZED VSLAM and OpenCV on a Jetson Orin Nano for autonomous navigation
- Created a hierarchical state machine in ROS2 to coordinate autonomous sub-systems, including target tracking, ball acquisition, and launching sequences
- Engineered a custom flywheel system and x-drive chassis using Fusion 360, including all fabrication and assembly

Technical Skills

Languages: Python, C++, Java
AI/Robotics/Sim: Isaac Sim/Lab/Gym, MuJoCo, PPO, PyTorch, 3DGS, SLAM, Behavior Trees, ROS2, Nav2, Rviz
Hardware: Jetson Orin Nano, Boston Dynamics Spot, Teensy 4.1, Arduino, Jackal UGV
Tools: Docker, Git, Linux (Ubuntu), Fusion 360